



Preoperative Planning, Diagnosis, and Staging for Brain Tumors

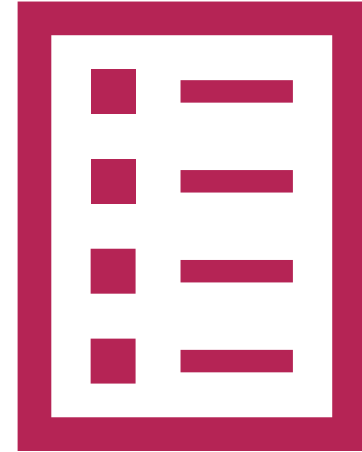
SPARK ACADEMY COURSES 2025

**ND A BANKOLE, MD, MSc
Neurosurgeon, PhD student**

**U1253 Ibrain, INSERM,
CHU DIJON, University of Tours, France**

PLAN

- **Introduction**
- **Understanding Brain Anatomy**
- **Clinical evaluation**
- **Diagnostic imaging modalities**
- **Preoperative surgical planning**
- **Grading for brain tumors**
- **Take home messages**
- **Conclusion and Q&A**

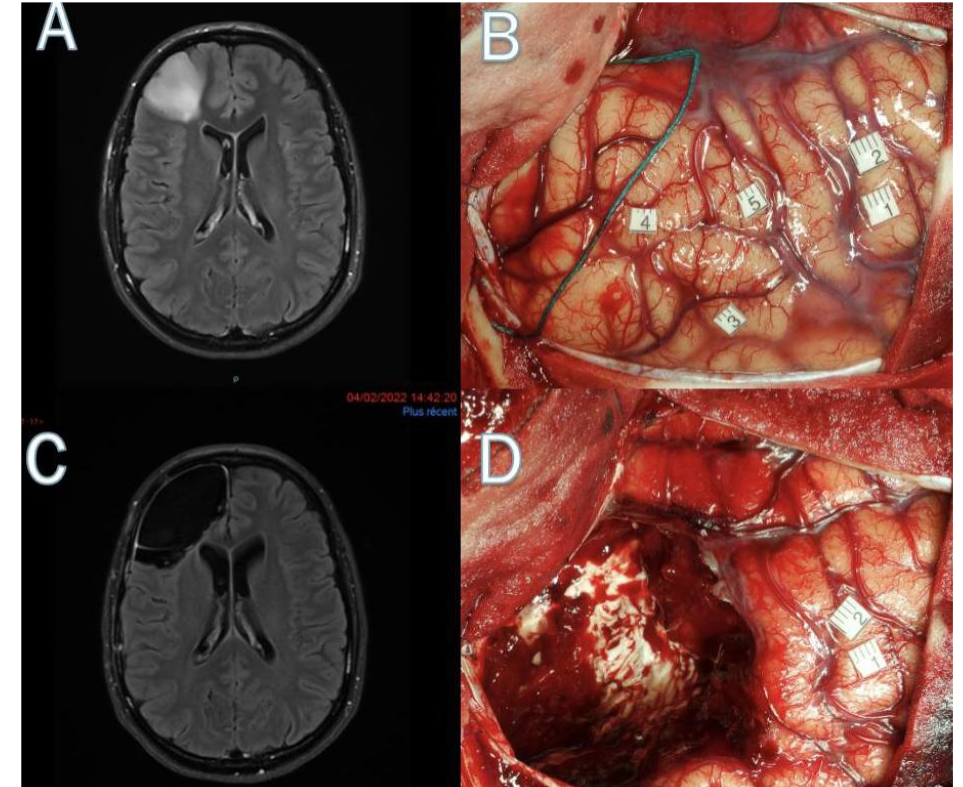


PLAN

- **Introduction**
- Understanding Brain Anatomy
- Clinical evaluation
- Diagnostic imaging modalities
- Preoperative surgical planning
- Grading for brain tumors
- Take home messages
- Conclusion and Q&A

INTRODUCTION

- Importance of preoperative planning
- Challenges in brain tumor management
- Goals: Maximize tumor removal, preserve function, guide therapy

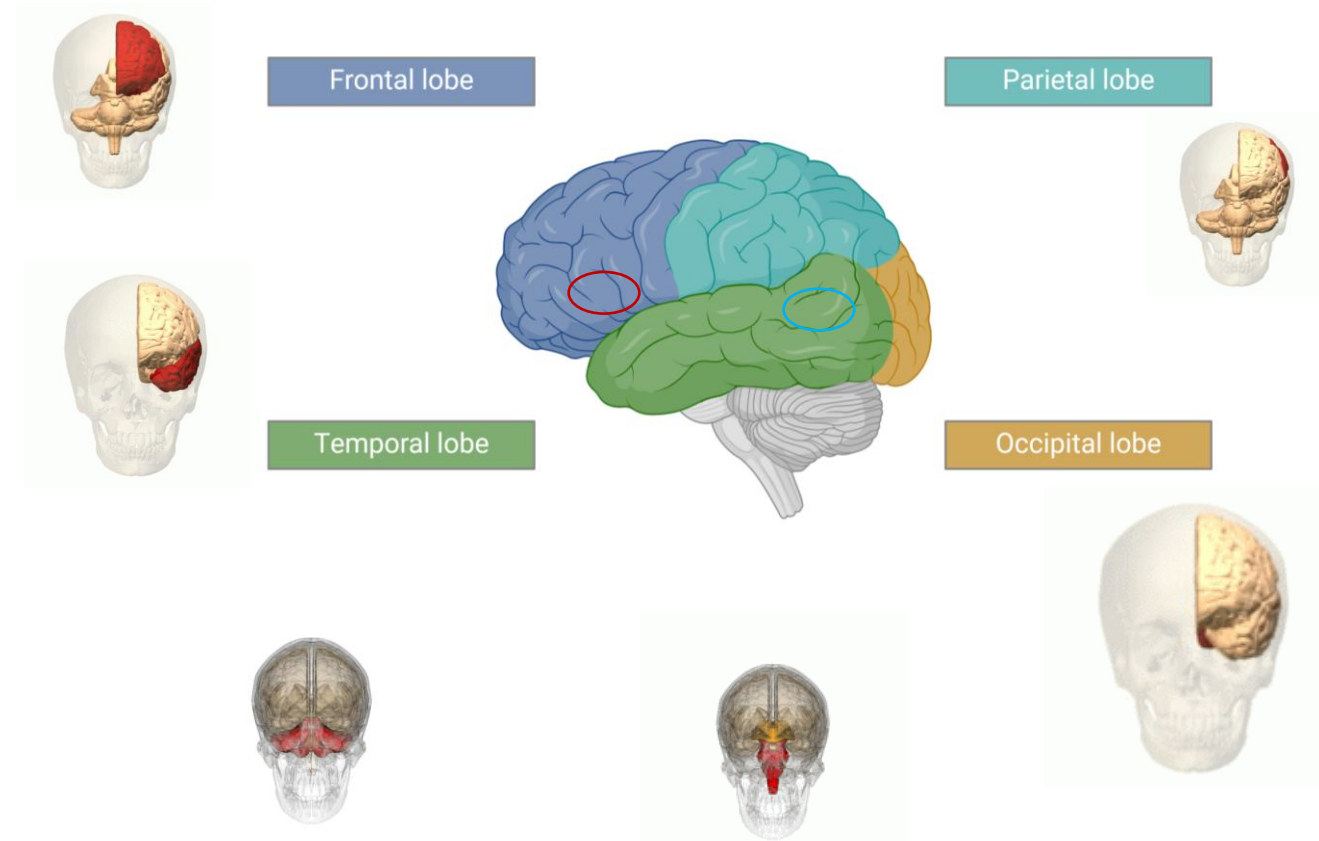
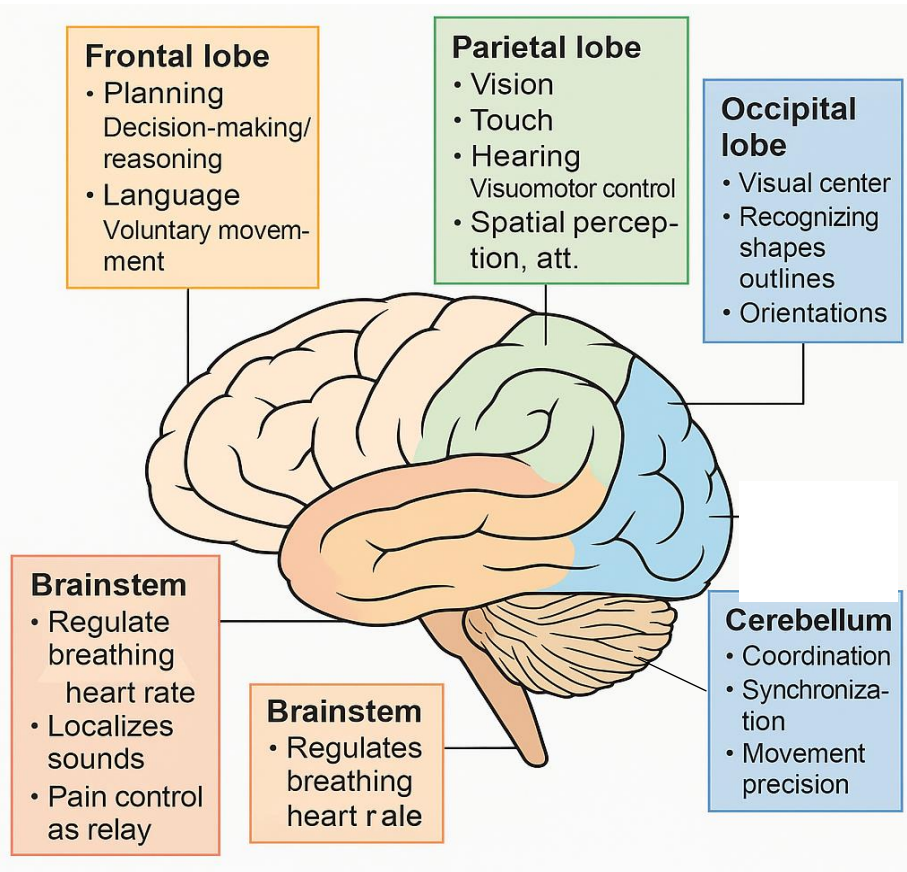


PLAN

- Introduction
- **Understanding Brain Anatomy**
- Clinical evaluation
- Diagnostic imaging modalities
- Preoperative surgical planning
- Grading for brain tumors
- Take home messages
- Conclusion and Q&A

BRAIN ANATOMY

Functionnal Organization: Essentials Functions



PLAN

- Introduction
- Understanding Brain Anatomy
- **Clinical evaluation**
- Diagnostic imaging modalities
- Preoperative surgical planning
- Grading for brain tumors
- Take home messages
- Conclusion and Q&A

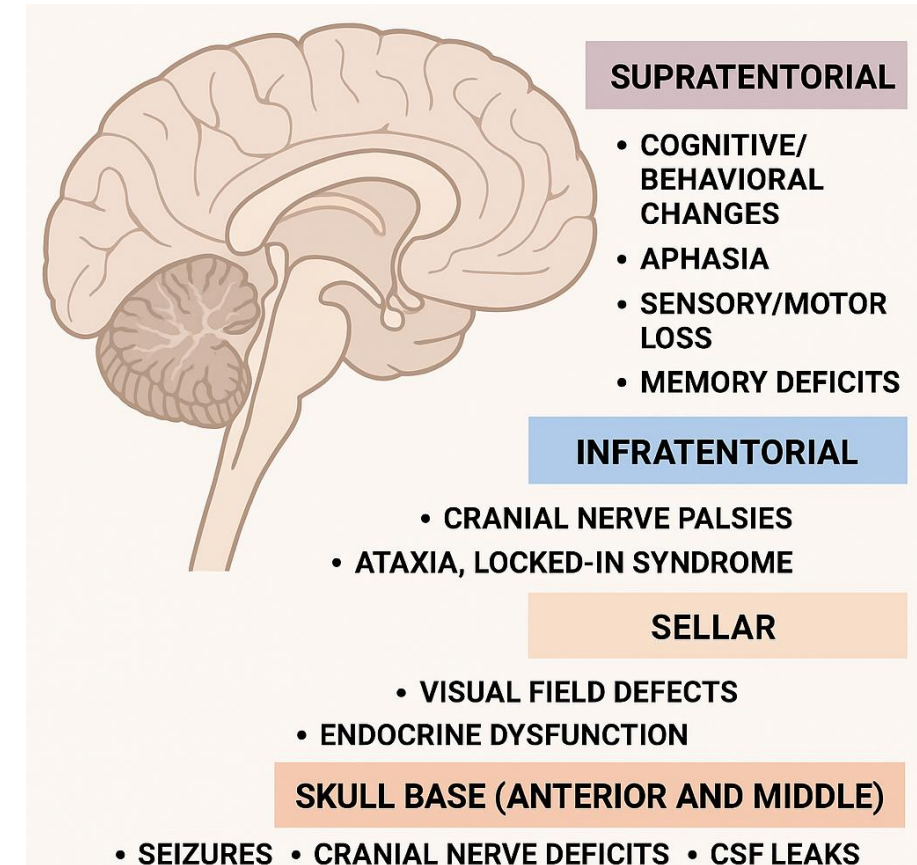
CLINICAL EVALUATION



Modified Rankin Score (mRS)



- Patient history and comorbidities, Age,
- History of symptoms installation (**suddenly, progressively+++**)
- Karnofsky status, independent prior event, mRs
- Neurological examination (GCS, neurologic deficit,
- Syndrome (summary of symptom, clinical sign)
- **Neuropsychological and cognitive test +++**
- Visual test, hearing test,
- Biology workup (hormonal dysfunction, inflammation, anemia...)

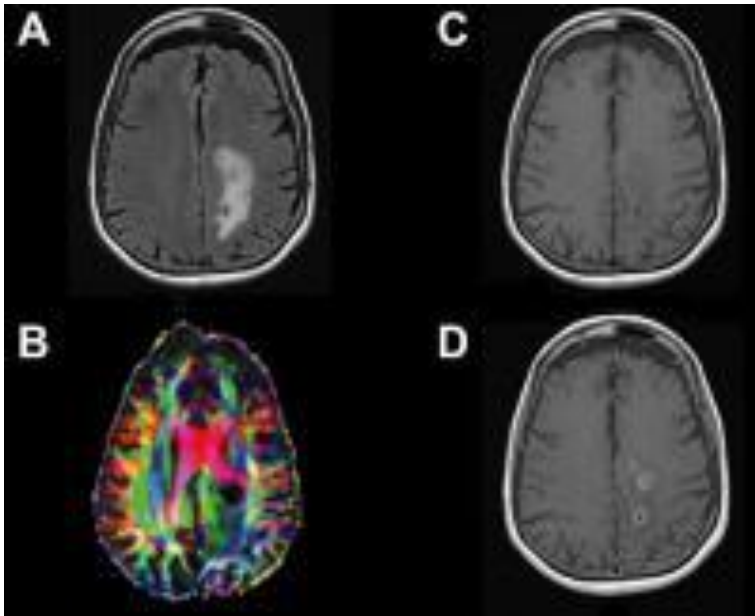


PLAN

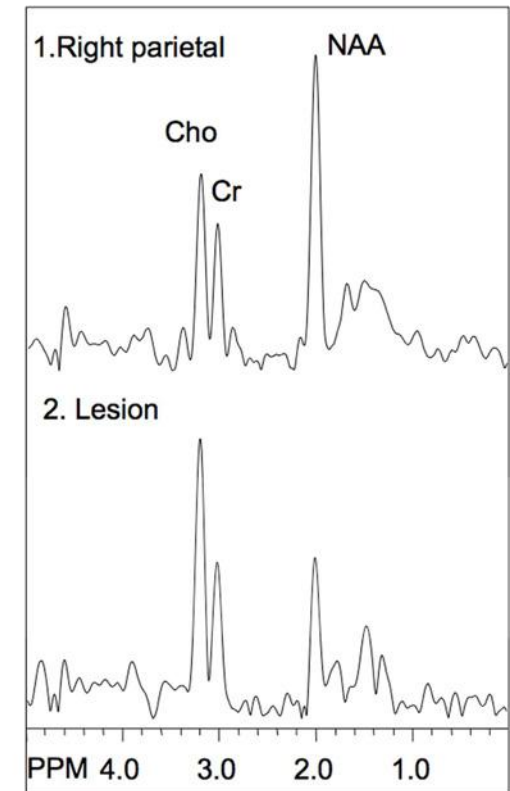
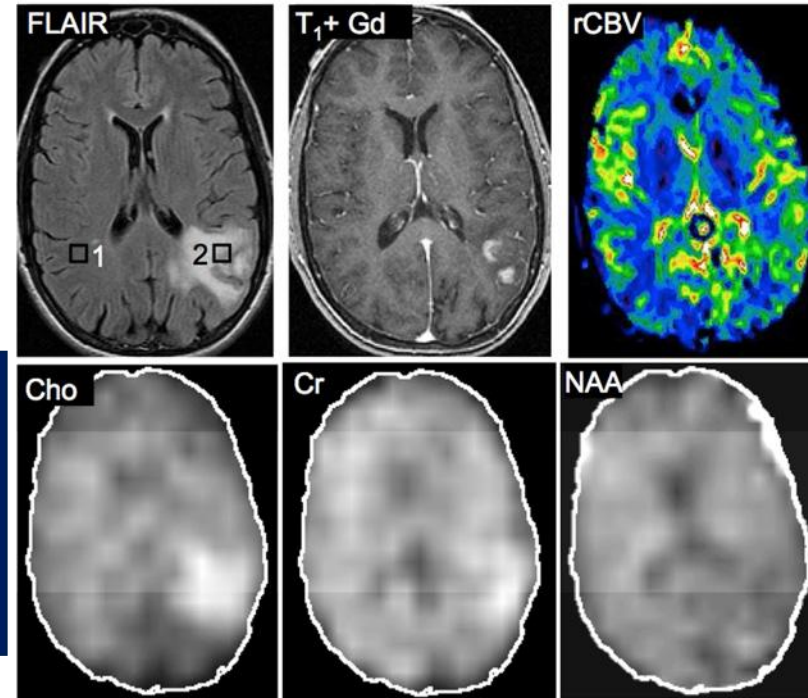
- Introduction
- Understanding Brain Anatomy
- Clinical evaluation
- **Diagnostic imaging modalities**
- Preoperative surgical planning
- Grading for brain tumors
- Take home messages
- Conclusion and Q&A

IMAGING MODALITIES FOR THE BRAIN

- **MRI (Gold standard):** T1, T2, FLAIR, Contrast-enhanced, DTI
- **CT Scan:** Quick evaluation, calcifications, hemorrhage
- **PET/SPECT:** Tumor metabolism
- **Advanced:** MR spectroscopy, perfusion, fMRI



Characteristics
of the tumors



> Rofo. 2023 Nov;195(11):989-1000. doi: 10.1055/a-2083-8717. Epub 2023 May 24.

Modern preoperative imaging and functional mapping in patients with intracranial glioma

Nico Sollmann ^{1 2 3 4}, Haosu Zhang ⁵, Christopher Kloth ¹, Claus Zimmer ^{2 3},
Benedikt Wiestler ^{2 6}, Johannes Rosskopf ^{1 7}, Kornelia Kreiser ^{1 8}, Bernd Schmitz ^{1 7},
Meinrad Beer ¹, Sandro M Krieg ^{3 5}

Review > Neuroimaging Clin N Am. 2010 Aug;20(3):293-310. doi: 10.1016/j.nic.2010.04.003.

Imaging of brain tumors: MR spectroscopy and metabolic imaging

Alena Horská ¹, Peter B Barker

PLAN

- **Introduction**
- Understanding Brain Anatomy
- Clinical evaluation
- Diagnostic imaging modalities
- **Preoperative surgical planning**
- Grading for brain tumors
- Take home messages
- Conclusion and Q&A

PREOPERATIVE SURGICAL PLANNING

- Matching symptoms, imaging, and pathology
- **Evaluation of Tumor resectability, Goal : GTR, STR, PR, Biopsy**
- Functional mapping: Language, motor areas
- **Risk stratification +++, Anesthesiology workup, AAA possibility**
- **surgery steps planification,**
- Intraoperative navigation planning
- **Importance of multidisciplinary team approach +++**
- **Preoperative Patient Counseling:**
Expected outcomes, Risks and benefits, Rehabilitation and recovery plan
- **Follow-Up Strategy:**
Immediate post-op imaging, Long-term surveillance, Recurrence management, Adjuvant therapies

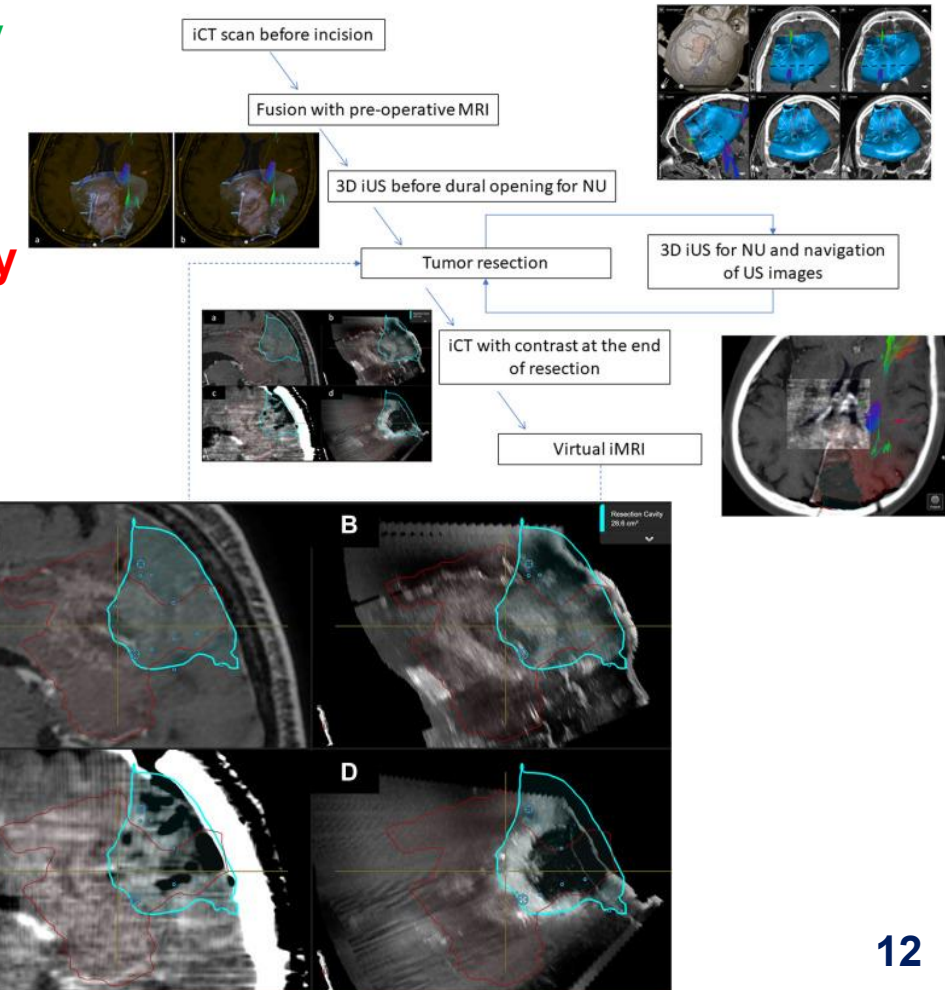
> World Neurosurg. 2022 Aug;164:330-340. doi: 10.1016/j.wneu.2022.05.133. CR Epub 2022 Jun 3.

Intraoperative Integration of Multimodal Imaging to Improve Neuronavigation: A Technical Note

Edoardo Mazzucchi ¹, Giuseppe La Rocca ², Patrick Hiepe ³, Fabrizio Pignotti ², Gianluca Galieri ², Domenico Policicchio ⁴, Riccardo Boccaletti ⁵, Pierluigi Rinaldi ⁶, Simona Gaudino ⁵, Tamara Ius ⁷, Giovanni Sabatino ²

Affiliations + expand

PMID: 35667553 CR DOI: 10.1016/j.wneu.2022.05.133 CR



PLAN

- Introduction
- Understanding Brain Anatomy
- Clinical evaluation
- Diagnostic imaging modalities
- Preoperative surgical planning
- **Grading for brain tumors**
- Take home messages
- Conclusion and Q&A

Grading for brain tumors

History: WHO 1979,1993,2000,WHO 2016, WHO 2021

Review > Neuro Oncol. 2021 Aug 2;23(8):1231-1251. doi: 10.1093/neuonc/noab106. CR

The 2021 WHO Classification of Tumors of the Central Nervous System: a summary

David N Louis¹, Arie Perry², Pieter Wesseling^{3,4}, Daniel J Brat⁵, Ian A Cree⁶, Dominique Figarella-Branger⁷, Cynthia Hawkins⁸, H K Ng⁹, Stefan M Pfister¹⁰, Guido Reifenberger¹¹, Riccardo Soffietti¹², Andreas von Deimling^{13,14}, David W Ellison¹⁵

Summary of WHO Classifications of Brain Tumors (2021, 5th Edition)

The 2021 WHO Classification of Tumors of the Central Nervous System (CNS) represents a major update, integrating molecular diagnostics with traditional histology to improve accuracy and prognostic value

Restructuring of Major Tumor Groups



Diffuse Gliomas

Now classified based on IDH mutation (IDH-mutant vs. IDH-wildtype) and 1p/19q codeletion status. Oligoastrocytoma and glioblastoma subtypes are redefined or eliminated, all IDH-mutant gliomas without 1p/19q codeletion are classified as astrocytomas



Medulloblastomas and Embryonal Tumors

Divided into molecular subgroups (e.g., WNT-activated, SHH-activated). CNS primitive neuroectodermal tumor (PNET) was removed as a category

+ New Entities

- + diffuse midline glioma, H3 K27M-mutant
- + RELA fusion-positive ependymoma
- + embryonal tumor⁺ with multilayered rosettes, C18MC-altered

GLIOMAS, GLIONEURONAL AND NEURONAL TUMORS

- Adult-type diffuse gliomas:
 - Astrocytoma, IDH-mutant (grades 2–4)
 - Oligodendroglioma, IDH-mutant, 1p/19q-codeleted (grades 2–3)
 - Glioblastoma, IDH-wildtype (grade-4)
- Pediatric-type low-grade and high-grade gliomas
- Circumscribed astrocytic gliomas
- Glioneuronal and neuronal tumors

CHOROID PLEXUS TUMORS

- Choroid plexus papilloma, carcinoma

MENINGIOMA

- Now a single tumor type with multiple subtypes, graded 1–3 within the type; molecular markers (TERT, CDKN2A/Bican indicate grade 3

EPENDYMAL TUMORS

- Classified by location (supratentorial, posterior fossa, spinal) and molecular features (e.g. ZFTA fusion, YAP1 fusion)

EMBRYONAL TUMORS

- Medulloblastoma (WNT-activated, SHH-activated, non-WNT/non-SHH, all grade 4)
- Other CNS embryonal tumors (e.g., ETMR, AT/RT, CNS neuroblastoma, CNS tumor with BCOR alteration)

OTHER GROUPS

- Pineal tumors
- Cranial and paraspinal nerve tumors
- Mesenchymal, nonmeningothelial tumors
- Melanocytic tumors
- Hematolymphoid tumors
- Germ cell tumors
- Sellar region tumors
- Metastases

Key Principles

Integrated Diagnosis: Combines histologic features with molecular markers for a layered, more precise diagnosis

Tumor Grading

Uses Arabic numerals (1–4) for grades and assigns grade within tumor types rather than across types

Adult-type vs. Pediatric-type

Clearly separates adult and pediatric diffuse gliomas and other tumor families based on molecular and clinical behavior

New Entities & Nomenclature

Introduces new tumor types and updates names to reflect molecular alterations

PLAN

- Introduction
- Understanding Brain Anatomy
- Clinical evaluation
- Diagnostic imaging modalities
- Preoperative surgical planning
- Grading for brain tumors
- **Take home messages**
- Conclusion and Q&A

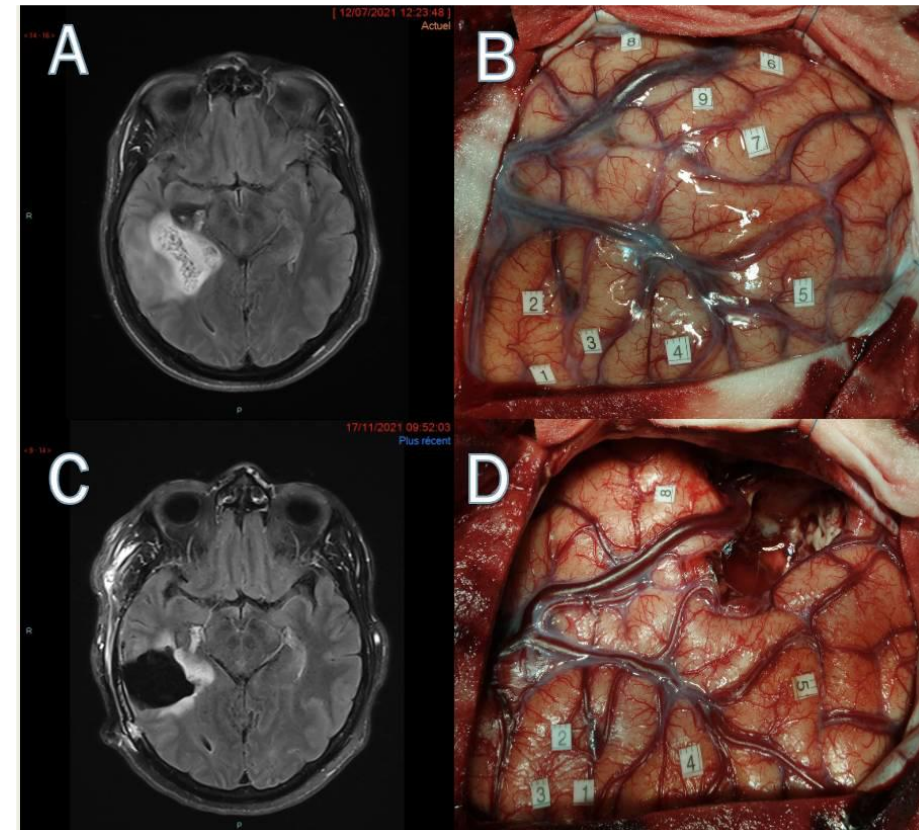
TAKE HOME MESSAGE

- **Clinical, biology workup is mandatory**
 - Correlation establishment with imaging, pathology
- **Imaging modalities : structural anatomy involved and characteristics of tumors**
- **Optimal planning of each steps of surgery with risk stratification (Anesthesiology consideration)**
- **Evaluated the possibility to be less Invasive (AAA vs. Hypnosis)**
- **Follow-up planification: Adjuvant therapies, recurrence according to the WHO 2021 grading of tumor**
 - through pathology, biomolecular workup, and Multidisciplinary concertation

> Neurochirurgie. 2023 Nov;69(6):101494. doi: 10.1016/j.neuchi.2023.101494. CR Epub 2023 Sep 14.

Asleep-awake-asleep versus hypnosis for low-grade glioma surgery: long term follow-up outcome

Nourou Dine Adeniran Bankole ¹, Ulrick Sidney Kanmounye ², Abdessamad El Ouahabi ³, Ilyess Zemmoura ⁴



PLAN

- Introduction
- Understanding Brain Anatomy
- Clinical evaluation
- Diagnostic imaging modalities
- Preoperative surgical planning
- Grading for brain tumors
- Take home messages
- **Conclusion and Q&A**

CONCLUSION

- Planning is essential in brain tumors surgery
- Clinical and Imaging correlation is crucial to be optimal in management
- Goal of surgery is to remove as so much possible the tumor without compromise functional outcomes
- **Follow-up is crucial and should be planned before surgery**
 - (WHO classification 2021)
- **Multidisciplinary concertation (neuro-oncologist-radiotherapist, neurosurgeon)**

Henry Kissinger: 'If
you don't know
where you are going,
every road will get
you nowhere.'



Thanks for your Kind Attention

Q&A ???



18/04/2026